

ADI-Autonomous Disaster Intelligence for Crowd Management and Disaster

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Abstract: *Managing crowds and responding to disasters is a crucial aspect of public safety, especially during major events, festivals, or natural disasters. The project ADI (Autonomous Disaster Intelligence) plans to develop a state-of-the-art quadcopter drone powered by machine learning and computer vision to manage crowds autonomously and respond to emergencies. The ADI drone utilizes RGB cameras, heat mapping, audio analysis, and autonomous navigation to detect human distress and abnormal crowd behaviour in real-time. The drone will operate offline using Raspberry Pi 5 and is designed to work even in areas where connectivity is challenging. The ADI will have an onboard public announcement system to provide guidance to crowds during an emergency. The new technology does not just increase the effectiveness of crowd monitoring, and disaster response, it also fills voids in existing traditional surveillance systems. Above all, ADI's goal is to enhance public safety, provide quick response in emergencies, and supply law enforcement and rescue teams vital real-time intelligence in an emergency.*

Keywords: *Crowd management, Disaster response, Quadcopter drone, Machine learning, Real-time monitoring, Audio analysis, Heat mapping, Behaviour Analysis.*

1. Introduction

As the world evolves rapidly, public safety issues during large-scale events or unexpected emergencies require innovative solutions. The ADI drones is a highly-capable, fully autonomous aerial system used for Search and Rescue (SAR) and crowd management in chaotic situations. This snapshot shows how the ADI drone utilizes cutting-edge technologies, including high-performance computing (e.g. Raspberry Pi 5) and deep learning objective and audio analysis, to deliver quality situational awareness and rapid emergency response continuously and reliably. Combining visual data, audio signals, real-time maps, and intelligent navigation, ADI will transform disaster response and crowd management.

2. Hardware Architecture and System Overview

At its heart, the ADI drone uses a robust and versatile computing platform. The system is powered by the Raspberry Pi 5 (4GB), which acts as an edge processor. This device is responsible for managing all real-time data processing, sensor fusion, and decision-making tasks—even in environments where network connectivity is unreliable or non-existent.

2.1. Processing Unit

Raspberry Pi 5 (4GB): The Raspberry Pi 5 is selected to be our primary processing unit, allowing the drone to operate autonomously from deep learning to integration of sensors. The memory capacity and processing speed are sufficient to ensure complex algorithms can run seamlessly, operating with no interruptions in remote disaster scenarios or at public gatherings.

2.2. Sensor Suite

ADI's sensor suite is one of its most critical components. The drone is outfitted with:

- **RGB Camera:** A camera that has high resolution and captures images in the visible light spectrum. It is important for operations that are conducted during daylight and integrates with deep learning models without interruption.
- **IR (Infrared) Camera:** This camera operates at 1080p and is designed for low-lux conditions. The IR camera turns on based on lux thresholding algorithms to provide human detection in the visible spectrum while detecting thermal contrast in the IR spectrum, ensuring it operates day and night.
- **Omnidirectional Microphones:** The omnidirectional microphones enable audio recording of the environment utilizing multiple microphones that can analyze audio patterns in real-time to potentially identify distress signals.
- **Additional Sensors for Navigation:** A combination of GPS, IMU, and barometric sensors all provide helpful information for localization. SLAM (Simultaneous Localization and Mapping) algorithms provide paths for AI-assisted exploration ensuring stable flight and accurate navigation in non-GPS environments.

2.3. Actuation and Communication Hardware

- **Quadcopter Frame & Motors:** A durable, 450-size quadcopter frame with 1400kv brushless motors ensures stability, thrust, and resilience in turbulent environments.
- **Onboard Public Announcement (PA) System:** This system allows for two-way communication and also broadcasts a variety of pre-recorded, or AI generated emergency instructions. Its inclusion on the ADI drone is vital for helping guide survivors during emergencies.
- **Secure Radio Telemetry:** Data intel i.e. real-time heat maps and sensor readings are transmitted back to ground control using secure radio telemetry channels. By securing connections, it ensures that wireless data with critical content, is relayed without delay, which contributes to emergency personnel making rapid decisions.

3. Operational Scenarios and Real-World Applications

The ADI drone is built to operate in a variety of unpredictable but reliable situations. Its autonomous flight capabilities and sensor capabilities make it a useful tool for disaster response in addition to crowd management.

3.1 Disaster response:

In a catastrophic event such as an earthquake, flood, or building collapse, the ADI drone is essential in saving lives. Its off-grid and disconnected environment capabilities means it can inherently search for victims and identify them even if communications are all down.

- **Victim recognition in extreme circumstances:** Combined with RGB and IR imaging using a deep-learning model, the ADI can detect, classify and recognize human postures that indicate injury or unconsciousness quickly. This is critical in a disaster where every second counts.
- **Audio-visual capabilities:** In case the drone is unable to see a person, due to debris or poor visibility, the audio will create alerts even if the person is too far from the drone for visual recognition.
- **Thermal heat mapping:** The system also produces heat maps that show indications of heat sources like body clusters and hazards like fire zones. This information can help ground rescue teams prioritize their search areas, and allocate resources.
- **Offline Operation:** The onboard Raspberry Pi 5 and integrated sensors allow the ADI drone to function independently of external networks. This offline capability is indispensable during natural disasters where infrastructure may be compromised.

3.2 Crowd Control and Public Safety Challenges

Crowd control and responses to emergencies or incidents present their own unique challenges at large events, festivals, sports events, political conventions, etc. Often traditional surveillance does not yield real-time situational awareness of crowd dynamics that may need to be monitored as crowds and situations change rapidly. The ADI drone will help address these limitations through automated, continuous monitoring and analysis.

- **Real-time Surveillance:** The ADI drone's high-definition cameras and real-time processing capabilities allow it to monitor large crowds for unusual behaviour or extremist activities.
- **Heatmap-Based Crowd Analysis:** The drone's proprietary heating mapping feature gives you visual cues regarding crowd density and movement patterns. Location heatmapping can assist in understanding and locating potential bottlenecks or high-risk crowd zones that may lead to an emergency.
- **Automated Notifications:** If the system detects unusual behaviour or circumstances such as sudden panic or irregular movements, you will automatically receive alerts to your ground control. Proactive communication allows incidents to be mitigated to prevent an increased crowd issue / incident from escalating to a disaster.

- **Public Announcement System:** In the event of an incident or emergency, the onboard PA system can be used to broadcast pre-recorded, or AI produced messages to the crowd to provide safety instructions, direction to a safe zone, or evacuation routes.

4. Integration of Technologies for Seamless Operations

The ADI drone's strength lies in its ability to integrate various technologies into a coherent, autonomous system. This integration ensures that the drone is not only capable of real-time detection but also of making informed decisions based on multi-source data.

4.1 Sensor and Data Fusion

- **Multi-Modal Data Collection:**
The simultaneous operation of RGB, infrared, and audio inputs presents a complete picture of the environment. The output of each sensor adds to each other, meaning that no important detail will be missed.
- **Robust Data Fusion Algorithms:**
Sophisticated algorithms fuse the data streams into a single real-time analysable dataset. It is the fused data that allows the system to detect subtle patterns or ongoing behaviour---perhaps a slight change in crowd dynamics or quickly developing panic.
- **Latency Reduction:**
By performing the processing of the data streams locally on the Raspberry Pi 5, the ADI drone tries to avoid latency in decision making. This quick action is particularly important in time-sensitive scenarios, such as quickly evacuating an area or responding to an escalating disturbance in a crowd.

4.2 Real-Time Processing and Decision Making

- **On-Board Processing:**
With all computational tasks performed on-device, the drone is not dependent on external networks. This autonomy ensures that critical operations continue uninterrupted, regardless of connectivity issues.
- **Autonomous Navigation and Obstacle Avoidance:**
The integration of SLAM-based AI navigation allows the drone to autonomously avoid obstacles and adjust its flight path in real time. This ensures safe operations even in cluttered or unpredictable environments.
- **Alert and Communication Protocols:**
When critical situations are detected—whether it's panic, fighting, or fallen victims—the system immediately triggers alerts. These alerts are transmitted securely to ground control and can also be accompanied by public announcements to guide affected individuals.

5. Deployment and Scalability

The ADI drone was designed to function as a scalable solution, allowing it to operate in a variety of conditions and operational requirements. With its modular design, the system can be enhanced and tailored for obstacles and applications established by those emerging technologies.

5.1 Scalability for Different Environments

- **Urban and Rural Environments:** In either heavily populated urban environments or rural settings, the ADI drone will provide reliability due to its extensive sensor suite and autonomous navigation. The system's ability to navigate complex structures and populated areas adds real value when deployed in urban environments, while in rural scenarios, its extended range and offline capabilities extend covered areas.
- **Large Scale Public Events:** Its ability to analyse crowd behaviour and gather real-time maps will make it ideally suited to monitor some back gather and sporting events. The detection of other modalities and the ability to move modalities as each challenge arises are ultimately more responsive than a single-sensor approach.
- **Disaster Situations:** Due to the autonomous and offline capable design of the ADI drone, the system can work when traditional communication has failed. The drone can operate totally independently without suffering a delayed impact from whether conditions or communication failure, allowing rescues to speed up and not delay when natural disasters occur or infrastructure fails and people require assistance.

5.2 Modularity and Potential for Future Updates

- **Hardware Upgradability:** The ADI system can integrate future hardware advancements due to its modular design. When sensor technology and processing units improve, upgrades will be made without fully updating the system architecture.
- **Software Adaptability:** Likewise, the software stack can be adapted. Improvements in deep learning frameworks, computer vision libraries, and audio processing algorithms can and will work to improve detection accuracy and operational efficiency within the ADI system.
- **Interoperability with Other Systems:** The ADI drone is developed to work alongside previous emergency response systems. Secure communication and data sharing capabilities make the drone able to communicate with other surveillance systems, command centres, and rescue coordination systems.

6. Advanced Crowd Behaviour and Emergency Detection

Apart from its central functions, the ADI drone project integrates a number of advanced detection priorities that are consistent with contemporary public safety needs. These priorities address the complexities of crowd behaviours and specified emergencies.

6.1 Panic & Running Detection:

One of the key priorities for the ADI project is distinguishing between ordinary walking behaviour and chaotic running. In several public safety situations, a sudden increase in running behaviour can be indicative of panic. The system is developed to:

Analyse Movement Patterns:

The system is utilizing video footage from high-resolution RGB and infrared cameras and uses deep learning models to observe patterns in individual and collective movement. This will allow a characterization of calm, orderly walking as opposed to abrupt chaotic running.

Trigger Emergency Protocols:

If the drone sees a big group of people running in an uncontrolled way, it marks that area as a potential hazard. This area would be queued as an area of interest, and potentially provide some alerts to rescue or law enforcement.

6.2 Crowd Behaviour Classification (Normal vs. Serious)

Beyond identifying running or panicked individuals, the ADI drone is equipped to classify the overall behaviour of a crowd. This capability is crucial for:

- **Differentiating Peaceful Gatherings from Distressed Crowds:**
The AI algorithms onboard will take instant readings of crowd density, movement flow, and body language and classify the crowd behaviour in two categories: Normal, Peaceful Movement vs. Distressed/Panic.
- **Contextual Analysis:**
The systems will not just look at raw data from the cameras and microphones, but also look back into historical norms and situational context to limit false alerts so that interventions are reserved for highly dangerous situations.
- **Prioritizing Intervention Areas:**
The classification of crowd behaviours will give authorities precise information about what area may require immediate engagement and what area is also relative safe.

6.3 Fight Detection

Physical altercations and fights can escalate into severe disturbances in crowded areas. The ADI drone addresses this challenge by:

- **Real-Time Monitoring for Physical Conflicts:**
Using high-quality video streams that are analysed with object detection approaches, the system learns to identify evidence of physical altercations. This could include rapid, aggressive body movements or clustering of people, which could indicate a fight.
- **Alert Generation:**
Once a fight is detected, the drone automatically generates an alert. The alert is accompanied by visual evidence and an accurate location, allowing law enforcement or security teams to respond quicker.

- **Integration with Audio Analysis:**

The system listens and detects for noise evidence, including yelling, aggressive language, or sounds of impact, that can help back up the video evidence and assist with fight detection.

6.4 Fallen/Injured Person Detection

In the wake of a stampede or any incident causing injuries, detecting individuals who have fallen is of paramount importance. The ADI drone's approach involves:

- **Posture Analysis via Computer Vision:**

Advanced object detection models are trained to identify when a person has fallen on the ground, either due to injury, fatigue, or the chaos of a stampede.

- **Thermal Imaging Support:**

The IR camera adds to the visual data by differentiating through thermal overlay. This is especially useful when individuals are obscured by dust, debris, or poor lighting conditions.

- **Rapid Response Enablement:**

Once a fallen or injured individual has been detected, the system identifies the incident location with a heatmap and sends that data to ground control, allowing rescue services an area to prioritize for assistance as soon as possible.

6.5 Authority Presence Detection (Police/Rescue)

To ensure efficient coordination during emergencies, the ADI drone is also capable of detecting the presence of authorities such as police or ambulances. This is achieved by:

- **Object Recognition for Vehicles and Uniforms:**

Machine learning algorithms can be trained to designate visual characteristics associated with emergency vehicles, police, and rescue personnel, identifying them as distinguished from civilian traffic.

- **Contextual Awareness:**

The system does not only detect the presence of public authorities, but also takes that information into consideration in its overall situational awareness. If, for example, police vehicles were located near an area identified for panic or a fight, the system could change the priority of alerts and provide a coordinated situational awareness capability.

- **Enhanced Communication with Command Centres:**

The detection of authority presence is captured and relayed to the Command Center along with other necessary information, allowing for coordinated situations. This will allow for rescue and law authorities to work in conjunction to remedy the situation.

7. Conclusion

The ADI project represents an innovative development in the field of public safety, crowd management, and disaster response. With a unique mix of state-of-the-art hardware - based on the Raspberry Pi 5 - and advanced software, it brings autonomous surveillance to a level never seen before, and adds real time autonomous decision-making capability. The trailing-edge software integrates a rope of sensors that combine both high resolution RGB and IR cameras with 6 omnidirectional microphones to detect the presence of people in distress across different environments. Moreover, the ADI drone's machine learning algorithms, including deep learning models for visual analysis and CNN-RNN hybrid models for audio detection, accurately identify changing contexts such as panic, fights, disabilities, fallen victims, and even the presence of emergency authorities. This is particularly important for both disaster and large public settings, where every second matters in the effort to mitigate risk and coordinate action. The ADI system is versatile, operating offline and with a SLAM based path planning function to autonomously navigate and orient itself. Detailed real-time heatmapping and secure radio telemetry serve as an added layer of support with actionable intelligence to emergency responders reducing time to action and potentially saving lives.

As public safety issues become more complex, the innovative blending of AI technologies, and autonomous navigation provides a new way of assessing and mitigating risk regardless of a natural disaster, a public event gone awry, or an urban emergency. The ADI drone provides a proactive and data-driven solution to both monitor and aid in creating a safe environment. Addressing key priorities (panic and running detection, crowd behaviour classification, fight detection, fallen/injured person detection, and authority presence detection) to meet contemporary public safety needs, the ADI drone is modular and designed for evolving needs and technology in order to retain continuing utility. The ADI project represents the start of a new era of intelligent drone-based capacity for public safety systems characterized by real-time situational awareness, rapid response, and collaboration in intervention.

This overview describes how the ADI drone is able to use new hardware and software technologies to provide an in-depth solution to large, public emergency events. The ADI drone, with its capability to utilize advanced visual and audio detection, autonomous navigation, and real-time processing, is poised to disrupt the future of public safety practices. The capabilities of the drone to detect subtle behaviour in crowd movement from a calm procession to running in panic, a physical fight, or a person down who may have fallen or become injured, in addition to the ability to recognize emergency responders, will allow the drone to manage ever-changing dynamics in the modern public space. Future work on the ADI system will increase precision and integration with more comprehensive emergency systems. Advancements in AI algorithms, sensor integration, and communication protocols will help to ensure that the ADI project will remain at the leading edge of intelligent drones. While challenges grow in communities throughout the world with emergencies from natural disaster events, mass public gatherings with the potential to become dangerous, or even an emergency response by law enforcement, the ADI drone will be a game-changer in maximizing public safety, response times, and ultimately saving lives.